

ALEXANDER LIQUE

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RJ, Brazil

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SKILLS

- Collaboration
- Adaptability
- Creativity
- Decision Making
- Self-Motivation

TECHNICAL SKILLS

- Python
- Scala
- C++
- Java
- Javascript
- SQL
- PL/pgSQL
- MongoDB
- PostgreSQL
- ChromaDB
- Matlab
- Pytorch
- Tensorflow
- Sklearn
- OpenCV
- Pandas
- Seaborn
- PowerBI
- Tableu
- Spark
- Hadoop
- NLTK
- RAG
- LangChain
- ONNX
- DVC
- MLflow
- React
- Node js
- FastAPI
- Odoo
- Jenkins
- Streamlit
- React N
- Flask
- Heroku
- AWS
- Docker

LANGUAGES

Spanish: Native

English: Intermediate

Portuguese: Intermediate

CERTIFICATES

Machine Learning with Python

Udemy

Specialized Program - Deep Learning

DeepLearning.AI

Data science

CTIC-UNI

ABOUT ME

I am currently pursuing a Master's degree in Decision Support Methods, focusing on Computational Intelligence. With a strong interest in research, I am particularly engaged in projects related to Data Science and Artificial Intelligence. I am always seeking opportunities to contribute meaningfully, take on new challenges, and advance in both academic and professional domains.

EDUCATION

University Student Pontifical Catholic University of Rio de Janeiro

- Mar. 2025 – at present
- Rj, Brasil
- I am undertaking a Master's program in Decision Support Methods, specializing in Computational Intelligence

University Student National University of Engineering

- Mar. 2019 – Jan. 2024
- Lima, Peru
- I earned my Bachelor's degree with a specialization in Computer Science, graduating in the top third of all students in the faculty.

Summer Studies Fundação Getulio Vargas

- Jan. 2024 – Feb. 2024
- Rio de Janeiro, Brazil
- Took courses at the School of Applied Mathematics at FGV, focusing on Machine Learning and Computer Graphics.

EXPERIENCE

Machine Learning Engineer | ICA PUC-Rio

- Jul. 2024 – Now
- Lima, Perú
- I currently work at the ICA Laboratory of PUC-Rio, collaborating closely with Petrobras to develop and implement artificial intelligence solutions for the oil and gas industry. My role involves research in deep learning, including the application computer vision and large language models (LLMs). I contribute to the design, development, and deployment of software systems that address complex challenges in this sector, working alongside the laboratory team on innovative projects.

Machine Learning Engineer | IMPACTO AI

- Jun. 2023 – Oct. 2023
- Lima, Perú
- In my previous role, I managed and developed chatbots (WhatsApp, Messenger, Instagram) using advanced technologies like ChatGPT and NLP. I also researched the latest AI models, such as Large Language Models (LLMs), to implement them in the company's solutions. Additionally, I contributed to projects involving OCR, image generation, Speech Recognition, and designed databases for the sales platform.

Software Developer | Ganemo

- Oct. 2022 – Abr. 2023
- Lima, Perú
- Software developer focused on testing and programming in the Odoo ERP, with experience in analysis, design, programming, testing and maintenance of business applications. Skills in Python, databases and containers, and use of agile methodologies such as Scrum and Kanban to ensure deadlines and software quality.

HONORS & AWARDS

ABET Project Fair and Competition | National University of Engineering

📅 Sep. 2022 📍 Lima, Perú

- First place in the project fair for ABET accreditation 2022-1 at the Faculty of Sciences, UNI, in the Computer Science category Group II (5th - 8th cycle). The project focused on studying transformer-based architectures to predict ratings in Yelp reviews.

Summer Program FGV EMap

📅 Jan. 2024 📍 RJ, Brazil

- I was selected as one of the ten students to receive a Grant-in-aid for Latin American students interested in participating in these courses at the FGV mathematics school. Rio de Janeiro, Brazil.

ACADEMIC INTERESTS

- Data Science
- NLP
- Computer Vision
- Machine Learning
- Deep Learning
- Evolutionary algorithm
- Reinforcement Learning

HOBBIES

- Reading Books
- Football (Soccer)
- Music
- Research

Software Developer Freelance | Smart City Lab

📅 Aug. 2022 – Apr. 2023 📍 Lima, Perú

- Developer and researcher at CTIC-UNI's SMART CITY, working on interdisciplinary projects in IoT, software, and artificial intelligence, as well as cloud computing. Proficient in various technology areas with practical experience applying technology to complex problems in different fields.

PROJECTS

Expected Goals Prediction Using Neural Networks

📅 Mar. 2025 – Jun. 2025

Developed a data-driven framework to predict football shot outcomes as expected goals (xG) using event-level data from StatsBomb. Applied a Genetic Algorithm to optimize preprocessing, feature selection, and neural network architecture. Achieved strong predictive performance with a Mean Absolute Error (MAE) of 0.0241 on an independent UEFA Euro 2024 test set. The model demonstrated high stability and generalization capability through residual analysis.

Semantic Segmentation for Soccer Fields

📅 Mar. 2025 – Jun. 2025

Developed and compared semantic segmentation models on the SoccerNet dataset, achieving the highest accuracy (mIoU 0.7036) using the U-Net architecture with the MIT-B0 encoder. The work balanced model performance and efficiency to enable real-time applications.

Deep Reinforcement Learning in Football Using PPO

📅 Mar. 2025 – Jun. 2025

Investigated the impact of architectural choices on PPO agents in the Google Research Football environment. Evaluated visual encoders, including MobileNetV2 and MiT, combined with temporal models such as LSTM and Transformer.

Rating Prediction in Reviews | 🐙 | 🌐

📅 Jan. 2022 – Sep. 2022

- I researched advanced natural language processing architectures, such as Transformers (BERT, ROBERTA, ELECTRA), for star rating classification on the YELP reviews dataset. I utilized Streamlit, PyTorch, and Python to explore and select the most suitable architecture.

EXTRACURRICULAR

Member ACECOM

📅 Mar. 2021 – Jan 2024 📍 Lima, Peru

- I was part of the board of directors of the association (Treasurer).
- I was a co-organizer for the realization of an exhibition of the Breca group at the Faculty of Science.
- I served as the head of the Artificial Intelligence area, where our focus was on promoting research in machine learning among members, as well as undertaking projects related to MLOps, data science, and data engineering.

Expositor ExpoScience FC

📅 Nov. 2023 📍 Lima, Peru

- Participation in the Faculty of Science ExpoScience with the aim of disseminating the research and activities conducted at the university to children from various schools. In this exhibition, my team and I presented the use of reinforcement learning in two agents competing for a ball. To achieve this, we utilized a Gym environment and neural networks with PPO.